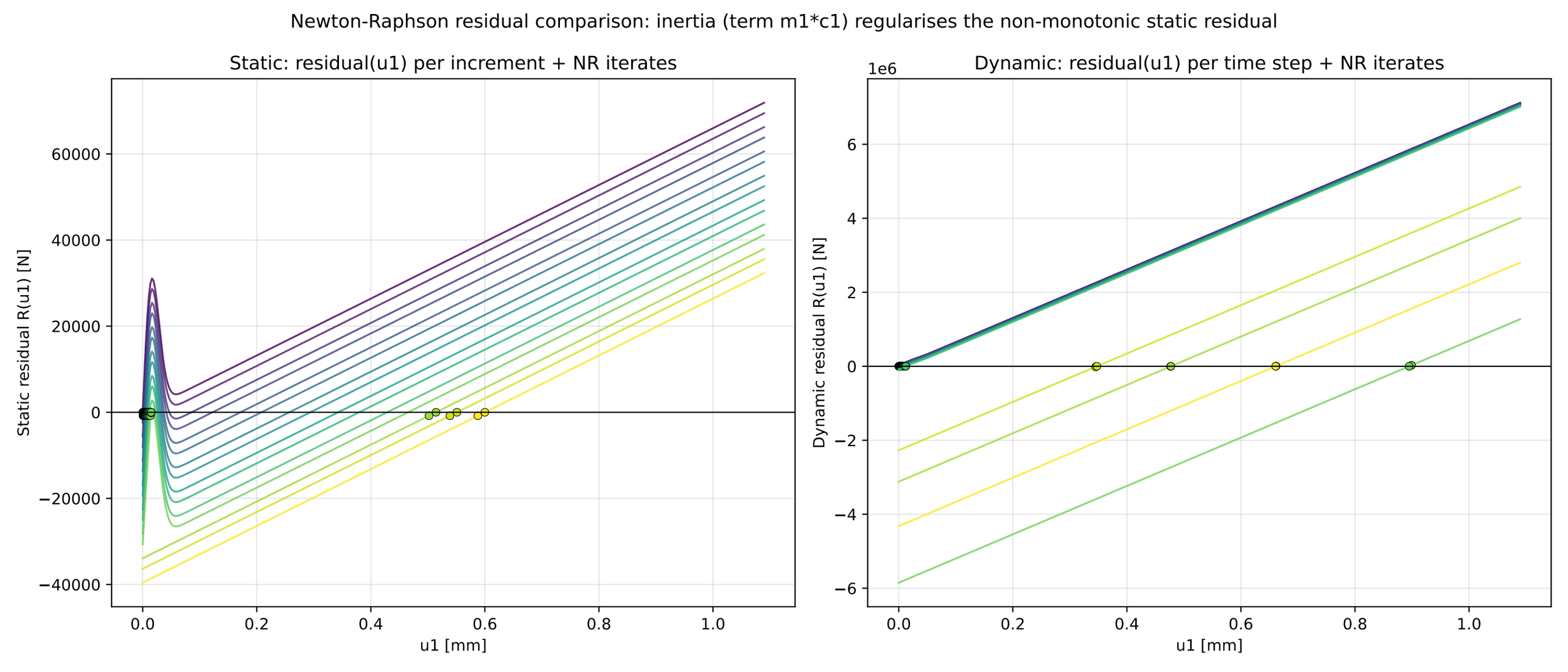
Newton-Raphson residual comparison: inertia (term m1*c1) regularises the non-monotonic static residual
Static: residual(u1) per increment + NR iterates
Static residual R(u1) [N]
60000
40000
20000
0
−20000
−40000
0.0
0.2
0.4
0.6
0.8
1.0
u1 [mm]
Dynamic: residual(u1) per time step + NR iterates
1e6
Dynamic residual R(u1) [N]
6
4
2
0
−2
−4
−6
0.0
0.2
0.4
0.6
0.8
1.0
u1 [mm]